Pain-Face: A Multimodal Platform Combining Facial Action-Unit Analysis with Remote Photoplethysmography for Automatic Pain Assessment

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Abstract. This report specifies Pain-Face, a real-time platform that estimates observable pain behaviour by combining frame-level facial action-unit (AU) analysis with remote photoplethysmography (rPPG) from the same camera. The work is a direct continuation of our earlier feasibility study on YOLOv8-based pain detection [1], and addresses the two gaps that study identified in its own limitations: that a detector may be **misled by "pain-like" expressions**, and that **incorporating biosignals could reinforce the model**. Pain-Face replaces binary pain/no-pain object detection with continuous estimation of the Prkachin and Solomon Pain Intensity (PSPI) score [2] over its full 0–16 range, adds a specificity correction that discounts the AUs that pain shares with smiling, and adds an autonomic channel derived from an eight-algorithm rPPG bank, fused so that facial evidence remains necessary.

We report validation of the facial channel on the Delaware Pain Database [3] against manual FACS coding and observer pain ratings (**n = 95** images). Our AU detection agrees with human coders better than OpenFace on the same images (mean F1 **0.629** vs 0.597; mean Cohen's κ **0.481** vs 0.381), and the resulting PSPI correlates with observer pain ratings at **r = +0.619** ($p = 2.3 \times 10^{-11}$), rising to **r = +0.649** with the specificity correction applied. On images whose dominant expression is **happy**, the correction reduces the score from 69.0% to **43.1%** of scale while pain images fall only from 89.3% to 83.9% — the correction discriminates rather than merely attenuating. We also report, with equal prominence, where the system fails: AU10 detection is near chance (F1 0.464, κ 0.076), disgust and anger remain substantially confounded with pain, and the correction slightly **reduces** categorical pain-vs-rest separation (AUC 0.846 → 0.828) even as it improves intensity correlation.

The rPPG and fusion channels are **verified but not validated**: still images cannot exercise them, and we state this explicitly rather than implying otherwise. A protocol for the required video-based validation is given.

Not a medical device. Pain-Face estimates observable pain **behaviour**, not what a person feels. It has not been clinically validated and must not be used for diagnosis, triage, or treatment decisions.

1 Introduction

1.1 Continuity with prior work

In [1] we demonstrated that a real-time object detector, YOLOv8, could classify facial images as pain or no-pain using AU-derived labels. Images were labelled by the presence of the PSPI action units (AU4, AU6, AU7, AU9, AU10, AU43) and thresholded at PSPI > 2. Across all classes the model reached mAP 0.893 at IoU 0.5, with precision 0.868 for **pain** and 0.919 for **no-pain**, and a peak F1 of 0.80.

That study closed with two explicit statements of what remained undone:

> "…it will be essential to determine how the model truly works in terms of interpretability and explainability. This step is crucial for determining whether the model accurately detects genuine pain **or can be misled by 'pain-like' expressions**. **Incorporating biosignals could help reinforce the model by integrating physiological data with facial expression analysis** to address this." — [1], Limitations

Pain-Face is the system built to answer both. The present report specifies it, measures it, and states precisely which of its claims are now evidenced and which are not.

1.2 Contributions

1. **A specificity correction for PSPI.** We quantify the extent to which PSPI over-reads non-pain expressions, because AU6 and AU10 are shared with smiling, and derive a correction that discounts the shared terms in proportion to smile evidence. On Delaware, this moves happy faces from 69.0% to 43.1% of scale while leaving pain faces essentially intact (§6).
2. **A fusion model in which facial evidence is necessary.** Autonomic arousal modulates the facial score multiplicatively through its remaining headroom, so that no degree of tachycardia produces a pain reading on a neutral face, and the index cannot saturate (§7).
3. **Abstention as a first-class behaviour.** The system refuses to report a pulse rate when the arriving frame rate would alias the cardiac band, refuses HRV on short windows, and distinguishes "no face" from "no pain" (§8). We measure the abstention boundary empirically (§10.2).
4. **A dual-rate real-time architecture** carrying 30 Hz colour and 6 Hz AU inference over one channel, with drop-not-queue backpressure (§4).
5. **Validation of the facial channel** against human FACS coding and observer pain ratings, including a three-way comparison with OpenFace, and a demographic breakdown (§9).

2 Why this system does not use YOLO

It is worth stating plainly, because the prior work did: **Pain-Face contains no YOLO model.** This is a deliberate methodological choice rather than an incidental one, and the reasoning matters.

YOLO is an object detector. It localises a region and assigns it a class confidence. Applied to pain, as in [1], this requires collapsing an ordinal construct into a binary label — in that study, PSPI > 2 — which discards the graded information PSPI was designed to carry. More importantly, a detection head returns a confidence score for a box. It cannot report **which** facial muscle group drove that score, because it never represents the AUs separately. That is exactly the interpretability gap the prior study identified.

Pain-Face instead estimates each action unit independently, computes PSPI analytically from those estimates over its full 0–16 range, and retains the four-term decomposition. This permits three things a detection head cannot offer: the continuous score, an explanation of which term produced it, and the detection of confounds that arise from specific shared terms — which is the mechanism of our specificity correction.

This is a change of task, not a claim that the earlier result was wrong. Binary screening and graded estimation are different problems; the YOLOv8 result stands for the problem it addressed. We note for completeness that YOLOv8 has since been superseded (YOLO11, 2024; YOLO26, 2026), but no version of YOLO is used here.

3 Background

3.1 FACS, PSPI, and the pain expression

The Facial Action Coding System decomposes facial movement into anatomically grounded action units. The Prkachin and Solomon Pain Intensity score [2] is the standard frame-level facial pain metric derived from it:

$$\mathrm{PSPI} = \mathrm{AU4} + \max(\mathrm{AU6}, \mathrm{AU7}) + \max(\mathrm{AU9}, \mathrm{AU10}) + \mathrm{AU43}$$

AU4 (brow lowerer), the orbital-tightening term (AU6 cheek raiser / AU7 lid tightener) and the levator term (AU9 nose wrinkler / AU10 upper lip raiser) each contribute 0–5; AU43 (eye closure) contributes 0–1. The score therefore ranges 0–16. Critically, the paired terms take the **maximum**, never the sum — AU6 and AU7 are alternative expressions of the same underlying orbital contraction.

Kunz and Lautenbacher [4] showed that pain expressions cluster into brow lowering, orbital tightening, levator contraction and open mouth, and — of direct relevance to this work — that what distinguishes pain from enjoyment is partly what is **absent**: AU12, the lip corner puller.

3.2 Remote photoplethysmography

rPPG recovers the cardiac pulse from subtle colour changes in skin caused by blood volume variation. We implement eight published algorithms: GREEN [5], ICA [6], PCA [7], CHROM [8], PBV [9], POS [10], LGI [11] and OMIT [12]. Heart-rate variability metrics follow the Task Force standard [13], and detrending follows Tarvainen [14].

4 System architecture

4.1 The dual-rate problem

The two channels have incompatible natural rates. Inter-beat interval estimation needs every frame at 30 Hz to resolve beat timing; the AU network sustains roughly 9–12 frames per second on the hardware measured here (§10.5). Running both at the slower rate would corrupt the pulse signal; running the detector at 30 Hz is not possible.

Pain-Face therefore runs them at independent rates over a single WebSocket:

- **Colour, every frame.** The browser computes mean skin RGB over forehead and both cheeks, taken from a MediaPipe face mesh with a YCrCb skin gate, and batches one second of triples per message.
- **Frames, a few times a second.** A 480 px JPEG is sent for AU inference at a configurable rate (default 6 Hz).

Frames arriving while an inference is in flight are **dropped, not queued**. Stale action units are worse than fewer of them, and dropping provides natural backpressure on slower hardware without unbounded latency growth.

4.2 Components

Module	Responsibility
`painface/pspi.py`	AU vocabulary, PSPI terms, the published formula, FACS-coder path
`painface/au.py`	Py-Feat wrapper: lazy init, NCHW tensor path, face quality
`painface/vitals.py`	Rolling rPPG buffer, autonomic feature reduction, abstention
`painface/fusion.py`	Specificity correction, baseline, fusion, trigger hysteresis
`painface/session.py`	Recording, episode statistics, JSON/CSV export
`painface/server.py`	FastAPI, `/ws/live`, offline video endpoint
`static/painface.html`	Capture, rendering, the operator interface

5 The facial channel

5.1 Detection

Action units are estimated with Py-Feat's `Detectorv2` [15], a multi-task network returning 20 AUs, 7 emotion probabilities, 68 landmarks, head pose, gaze and 52 blendshapes per detected face. Frames are supplied as NCHW `uint8` tensors, which avoids file I/O in the live path.

Two implementation constraints are worth recording. First, the available AU set does not include **AU27**; the open-mouth component must therefore rest on AU25 and AU26. Second, Py-Feat reports AU **occurrence probabilities** in $[0, 1]$, not FACS A–E **intensities** in $[0, 5]$. We map probability to each term's ceiling, which produces a score commensurate with the published 0–16 range, but a probability of 0.8 is not equivalent to a coder scoring intensity D. This is a documented approximation and a stated limitation (§12). Where true coder intensities are available, `pspi_raw()` computes the literal score with no approximation.

5.2 The specificity problem

PSPI is a valid measure of the pain expression, but it is not pain-**specific**. AU6 appears in the Duchenne smile; AU10 appears in both smiling and disgust. A face expressing enjoyment therefore accumulates PSPI points through terms it shares with pain.

This is not hypothetical. On Py-Feat's own bundled sample image — a face the model itself classifies as 99% **happy** — raw PSPI evaluates to **7.24 / 16**, or 45% of scale. Any system reporting that number as a pain intensity is reporting a smile as moderate pain.

5.3 The correction

Pain is separated from enjoyment by the absence of AU12 and by the brow: pain lowers it, happiness does not [4]. We define smile evidence

$$\sigma = \mathrm{clip}\big(\mathrm{AU12} \cdot \big(0.55 + 0.45 \cdot p_{\mathrm{happy}}\big) \cdot (1 - 0.7 \cdot \mathrm{AU4})\big)$$

and discount only the terms that smiling shares with pain, in proportion:

$$\mathrm{PSPI}_c = \sum_{t \in T} \mathrm{pts}_t \cdot (1 - s_t \cdot \sigma)/16$$

where $s_t = 0.85$ for the orbital term, 0.65 for the levator term, and 0 for brow lowering and eye closure, which smiling does not produce.

The literal PSPI is retained and reported alongside the corrected score. Both appear in the interface; neither is hidden. §9.3 measures whether the correction does what it claims on real faces.

6 The autonomic channel

Pain produces sympathetic activation, and the same camera that reads the face can measure it. This section specifies the full signal chain from pixels to heart rate, since that path determines what every downstream autonomic feature is worth.

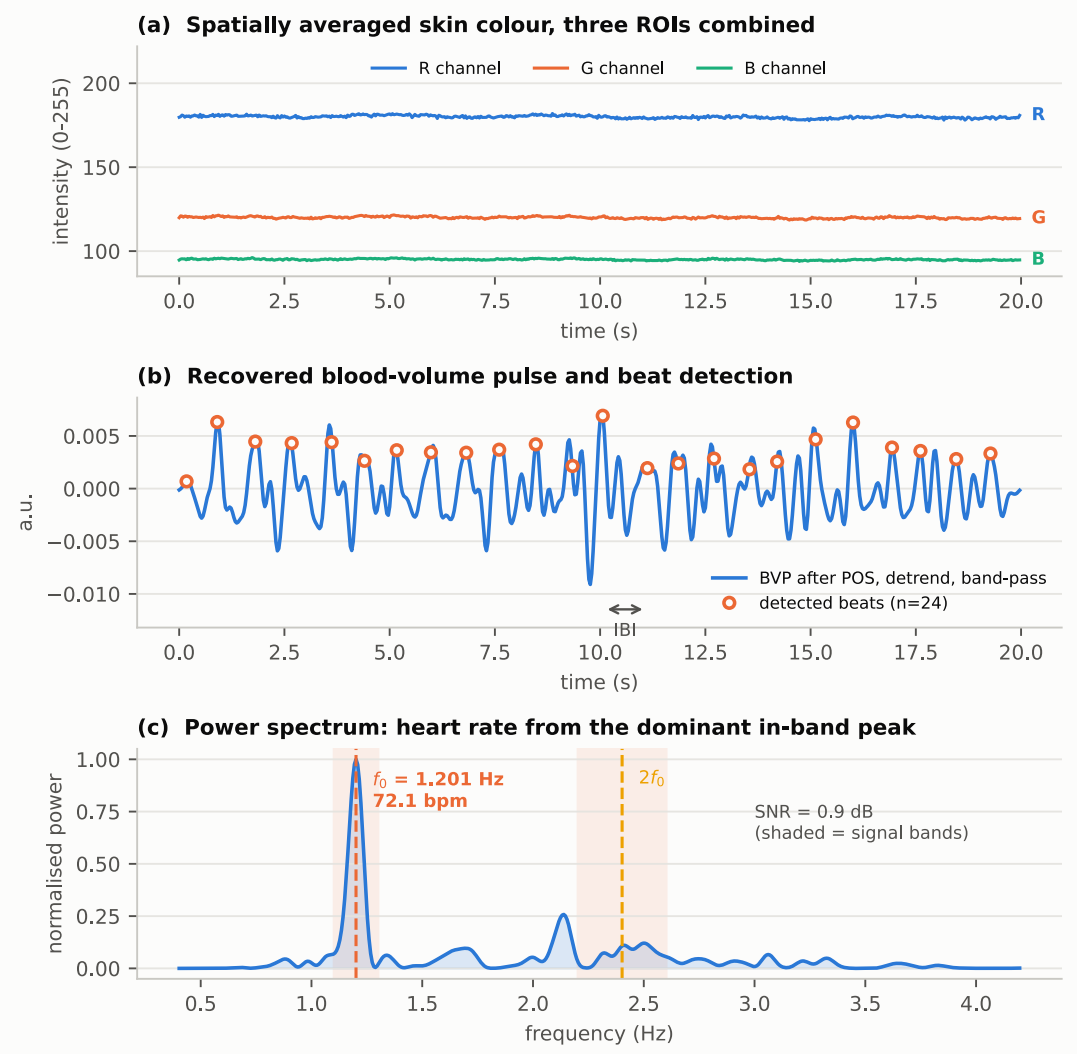


Figure 1: The rPPG chain on a synthetic trace of known rate (72.0 bpm). (a) spatially averaged skin colour; (b) the recovered pulse after projection, detrending and band-passing, with detected beats and one inter-beat interval marked; (c) the power spectrum, with the estimate taken from the dominant in-band peak at f_0 and the de Haan SNR bands shaded. Recovered rate 72.07 bpm.

6.1 Stage 1: region selection and skin segmentation

Three regions are tracked from the MediaPipe 478-point face mesh: the forehead and both cheeks, taken as convex hulls of fixed landmark sets. They are chosen for dense superficial microvasculature and relative freedom from hair, eyebrow and nasolabial shadow.

Within each hull, pixels are admitted only if they fall in the Chai and Ngan chrominance range for skin,

$$133 \le C_r \le 173, \quad 77 \le C_b \le 127$$

computed per pixel from the RGB triple. Hair, lips, eyes and background are thereby excluded; they carry no pulse and would otherwise attenuate it.

6.2 Stage 2: spatial averaging

For frame n and channel $c \in \{R, G, B\}$, each region k yields a mean over its admitted pixel set $\Omega_k(n)$, and the three regions are combined by pixel count so that larger, better-conditioned regions dominate:

$$C_c(n) = \frac{\sum_k \sum_{p \in \Omega_k(n)} I_c(p, n)}{\sum_k |\Omega_k(n)|}$$

Spatial averaging over 10^3–10^4 pixels is what makes the measurement possible at all: the per-pixel pulsatile modulation is far below camera noise, and averaging reduces that noise by roughly $\sqrt{N}$ while the pulse, being spatially coherent, is preserved.

6.3 Stage 3: uniform resampling

Browser frame timestamps are irregular. The three traces are sorted, duplicate timestamps dropped, and linearly interpolated onto a uniform grid at $f_s = 30$ Hz. Every subsequent stage assumes uniform sampling; §8 describes what happens when the arriving rate cannot support it.

6.4 Stage 4: temporal normalisation

Each channel is divided by its own temporal mean over the window,

$$C_n(t) = C(t)/\bar{C}$$

which removes the dependence on absolute illumination and skin tone, leaving the relative modulation that carries the pulse. The chrominance methods below operate on C_n in short sliding windows (1.6 s, Hann-weighted, 50 % overlap, overlap-added), so that slow illumination drift cannot bias the projection.

6.5 Stage 5: pulse extraction, eight algorithms

Each method reduces the three normalised channels to one scalar waveform. They differ in how they suppress motion and illumination artefacts, which share the colour subspace with the pulse.

GREEN [5] takes the green channel alone, where haemoglobin absorption is strongest: $S = G/\bar{G}$.

CHROM [8] forms two chrominance signals from a standardised skin-tone assumption and combines them so that the specular motion component cancels:

$$X_s = 3R_n - 2G_n, \quad Y_s = 1.5R_n + G_n - 1.5B_n, \quad S = X_s - (\sigma(X_s)/\sigma(Y_s))Y_s$$

POS [10] projects onto a plane orthogonal to the specular direction and applies the same alpha-tuning:

$$S_1 = G_n - B_n, \quad S_2 = G_n + B_n - 2R_n, \quad S = S_1 + (\sigma(S_1)/\sigma(S_2))S_2$$

PBV [9] uses the blood-volume signature p_{bv}, the normalised vector of per-channel standard deviations, to find the weighting that best explains the observed covariance:

$$W = (C_z C_z^T + \varepsilon I)^{-1} p_{\mathrm{bv}}, \quad S = (C_z^T W)/(p_{\mathrm{bv}} \cdot W)$$

where C_z is the mean-centred normalised matrix.

LGI [11] removes the dominant subspace, which carries illumination change, via the singular value decomposition $X = U\Sigma V^T$:

$$S = \left[(I - u_1 u_1^T)X\right]_G, \quad u_1 = \text{first column of } U$$

OMIT [12] achieves the same orthogonalisation through a QR decomposition, which is cheaper and better conditioned: $S = \left[(I - q_1 q_1^T)X\right]_G$.

PCA and **ICA** [6], [7] are blind source separation. PCA takes the eigenvectors of the channel covariance; ICA uses symmetric FastICA with the logcosh contrast, whitening first by eigendecomposition. Both produce three candidate components, and neither orders them by physiological relevance, so the component is chosen by in-band spectral concentration,

$$\rho(x) = \max_{f \in B} P(f) / \sum_{f \in B} P(f), \quad B = [0.7, 3.5] \text{ Hz}$$

selecting the candidate whose power is most concentrated at a single plausible cardiac frequency.

6.6 Stage 6: detrending

Extracted waveforms carry low-frequency drift from illumination and slow head motion. We apply smoothness-priors detrending [14], which removes a trend defined by a second-difference penalty rather than a fixed polynomial:

$$z_{\text{trend}} = (I + \lambda^2 D_2^T D_2)^{-1} z, \quad z_{\text{stat}} = z - z_{\text{trend}}$$

with $\lambda = 100$ and D_2 the second-order difference operator. The system is sparse and solved directly. Unlike a high-pass filter, this introduces no phase distortion at the band edge, which matters because beat timing is extracted downstream.

6.7 Stage 7: band-pass filtering

A third-order Butterworth band-pass restricted to $[0.7, 3.5]$ Hz (42–210 bpm) is applied **zero-phase**, forward and backward, in second-order-section form. Zero-phase filtering is not a cosmetic choice: a causal filter would shift every beat by a frequency-dependent delay and corrupt the inter-beat intervals on which all HRV metrics depend.

6.8 Stage 8: spectral estimation and heart rate

The heart rate is taken from the dominant spectral peak inside the cardiac band. Two estimates are computed.

The **periodogram** estimate applies a Hann window and zero-pads to at least 4096 points. Zero-padding does not add information, but it interpolates the spectrum so the peak can be located more precisely. The peak bin k is then refined to sub-bin resolution by fitting a parabola through the three surrounding points in the log-power domain, which is the correct interpolant for a Gaussian-shaped windowed peak:

$$\delta = \left(\frac{1}{2}\right) \frac{\ln P_{k-1} - \ln P_{k+1}}{\ln P_{k-1} - 2 \ln P_k + \ln P_{k+1}}, \quad f_0 = f_k + \delta \cdot \Delta f$$

$$\mathrm{HR} = 60 f_0 \quad [\text{bpm}]$$

Without this refinement the resolution would be limited to $\Delta f = f_s / N_{\text{fft}}$; with it, the error on a clean synthetic signal falls below 0.1 bpm (§10.3).

A **Welch** estimate (8 s segments, 50 % overlap) is computed in parallel. It trades frequency resolution for variance reduction, and large disagreement between the two is itself a signal-quality warning.

6.9 Stage 9: signal quality

Each method's output is scored with the signal-to-noise ratio of de Haan and Jeanne [8], which treats power at the fundamental and its second harmonic as signal and everything else in $[0.5, 4]$ Hz as noise:

$$\mathrm{SNR} = 10 \log_{10} \left(\frac{\sum_{f \in \mathcal{S}} P(f)}{\sum_{f \in \mathcal{N}} P(f)} \right)$$

$$\mathcal{S} = \{ f : |f - f_0| \le 0.1 \vee |f - 2f_0| \le 0.2 \}, \quad \mathcal{N} = [0.5, 4] \setminus \mathcal{S}$$

The second harmonic is included because the arterial pulse is not sinusoidal: the dicrotic notch places real physiological power at $2f_0$, and excluding it would penalise a good signal.

6.10 Stage 10: fusion across methods

The eight estimates are combined by a **weighted median** rather than a mean, with weights $w_i = \max(\mathrm{SNR}_i, -10) + 10$. The median is chosen deliberately: when one method locks onto a motion artefact its estimate is not merely noisy but wrong by tens of bpm, and a mean would be dragged by it while a weighted median is not. The weighting lets confident methods decide among themselves.

6.11 Beat detection and inter-beat intervals

Spectral HR gives an average rate; HRV requires individual beat times.

The waveform is re-filtered into an HR-adaptive band $[0.6 f_0, 1.6 f_0]$, which suppresses the second harmonic so the dicrotic notch is not detected as a beat. After z-scoring, peaks are found subject to a refractory distance of $0.6 \times 60 / \mathrm{HR}$ seconds and a prominence floor of 0.35. Any pair closer than 70 % of the median interval is

merged, keeping the taller. Each retained peak time is then refined by the same parabolic interpolation used in the spectrum, giving sub-sample timing.

Intervals are accepted only within physiological range, $0.28 < \mathrm{IBI} < 1.6$ s, and within 20 % of the rolling median. The rejected fraction is reported as `artifact_pct`; it is a quality figure in its own right.

6.12 HRV metrics

From the accepted interval series the standard time-domain measures are computed: SDNN, RMSSD, SDSD, pNN50 and pNN20, together with the Poincaré descriptors $\mathrm{SD1} = \sqrt{1/2}\sigma(\Delta\ \mathrm{RR})$ and $\mathrm{SD2} = \sqrt{2\sigma^2(\mathrm{RR}) - \mathrm{SD1}^2}$, and the Baevsky stress index $\mathrm{AMo}\ /(2\ \mathrm{Mo} \cdot \mathrm{MxDMn})$ on 50 ms bins.

Frequency-domain measures require an evenly sampled tachogram, so the intervals are cubic-spline interpolated to 4 Hz and a Welch spectrum is integrated over the Task Force bands [13]: VLF 0.0033–0.04 Hz, LF 0.04–0.15 Hz, HF 0.15–0.4 Hz, yielding LF/HF and normalised units.

6.13 Respiration

Breathing modulates the pulse in three ways, and all three are estimated and median-fused. **RIIV**, baseline intensity modulation, is recovered by detrending the normalised green channel with a longer prior ($\lambda = 300$) and band-passing to 0.1–0.5 Hz. **RIAV**, pulse amplitude modulation, is taken from the envelope of detected beat amplitudes resampled to 4 Hz. **RSA**, respiratory sinus arrhythmia, is the dominant peak of the tachogram spectrum within the same band. Agreement between three mechanisms is stronger evidence than any one.

6.14 Perfusion and the SpO$_2$ ratio

The perfusion index is the pulsatile fraction of the green channel, $\mathrm{PI} = 100 \cdot 2\sigma(\mathrm{AC})/\ \mathrm{DC}$, which falls under vasoconstriction. The ratio-of-ratios $R = (\mathrm{AC}/\mathrm{DC})_R/(\mathrm{AC}/\mathrm{DC})_B$ is computed and displayed as an **uncalibrated index only**: without a reference oximeter to fit the calibration curve it is not an oxygen saturation, and the interface labels it as such.

6.15 Autonomic features and baseline normalisation

Six features are passed to the fusion model, each scored against the subject's own calm baseline, because absolute heart rate carries no information about pain while a rise above **this person's** rest does.

Feature	Direction	SD floor	Rationale
Heart rate	↑	2.0 bpm	tachycardia under nociception
RMSSD	↓	5.0 ms	parasympathetic withdrawal
SDNN	↓	5.0 ms	overall variability reduction
LF/HF	↑	0.3	sympathovagal shift
Respiration rate	↑	1.5 br/min	respiratory acceleration
Perfusion index	↓	0.15 %	peripheral vasoconstriction

Each is converted to a signed z-score against the baseline, with a minimum standard deviation (the "SD floor") so that an unusually steady rest period cannot magnify ordinary jitter into a large deviation. Scores are saturated and averaged to an arousal value in $[0, 1]$ where 0.5 denotes "at baseline". Until the baseline holds sufficient samples the channel reports `ready = false` and contributes nothing.

7 Multimodal fusion

7.1 The model

Let f be the corrected facial pain evidence in $[0, 1]$ and $d \in [-1, +1]$ the autonomic deviation from baseline. The fused intensity is

$$I = f \cdot (1 + k \cdot d \cdot (1 - f)), \quad k = W_a/W_f = 0.25/0.75 = 1/3$$

reported on a 0–10 scale for familiarity with numeric rating scales.

7.2 Two properties this form guarantees

Facial evidence is necessary. At $f = 0$ the expression evaluates to zero for every value of d. No degree of autonomic arousal can produce a pain reading from a neutral face. Measured: a neutral face at 130 bpm with suppressed HRV yields **NRS 0.2**. Arousal corroborates; it never originates.

The index cannot saturate. Modulation acts on the headroom $(1 - f)$ that remains, so $I \to 1$ asymptotically. A severe episode retains resolution near the top of the scale instead of pinning flat. An earlier additive formulation clamped a facial score of 0.76 to a maximum reading; this form returns 0.82 for the same inputs. Figure 7 shows the full response surface.

7.3 Episode segmentation

A latching trigger with hysteresis converts the continuous index into discrete episodes: the index must hold above an onset threshold for a **dwell** period to fire, and fall below a lower threshold for a **release** period to clear. A momentary grimace therefore does not start an episode, and a blink does not end one. Defaults: onset 0.30, release 0.18, dwell 0.6 s, release 1.5 s.

8 Abstention: refusing to report

A measurement instrument that reports a plausible wrong number is more dangerous than one that reports nothing. Three abstention rules are implemented.

Aliasing. The cardiac band extends to 3.5 Hz. A colour trace arriving below approximately twice that rate folds energy into the band and yields a confident, incorrect heart rate. Browsers throttle hidden tabs to roughly 1 fps, and loaded machines drop frames, so the arriving rate is measured rather than assumed. Below **7.7 Hz** (2.2 × the band ceiling) heart rate and HRV are withheld with a stated reason. This rule was added after observing the system report 52 bpm from a 1 Hz trace. §10.2 measures the resulting boundary.

Short windows. RMSSD and SDNN are unstable on short records. They are withheld, and excluded from the baseline, until the analysis window reaches 30 s.

No face. Py-Feat returns a populated row with `FaceScore = 0` and all AUs `NaN` when no face is present. Treating that as a neutral expression would score it as “no pain” rather than “no measurement”. The case is detected explicitly.

9 Validation: Delaware Pain Database

9.1 Protocol

The Delaware Pain Database [3] provides posed and evoked facial expressions with two forms of ground truth distributed alongside the images: manual FACS coding by trained human coders, and continuous observer pain ratings. OpenFace’s AU codes for the same images are also distributed, which permits a three-way comparison without us re-running that tool.

We scored every image for which manual coding and an image file could both be resolved: **n = 95**. AU probabilities were binarised at **0.5**, the same threshold the live system uses, so the reported agreement reflects deployed behaviour rather than a tuned operating point. Expression categories in the set: pain 45, disgusted 14, surprised 11, happy 9, sad 9, angry 6, fearful 1.

Scope. These are still images. They validate the **facial channel only**. The rPPG channel, the autonomic features, the fusion model and the episode trigger are not exercised by this data and remain verified but unvalidated (§11).

9.2 Agreement with human FACS coders

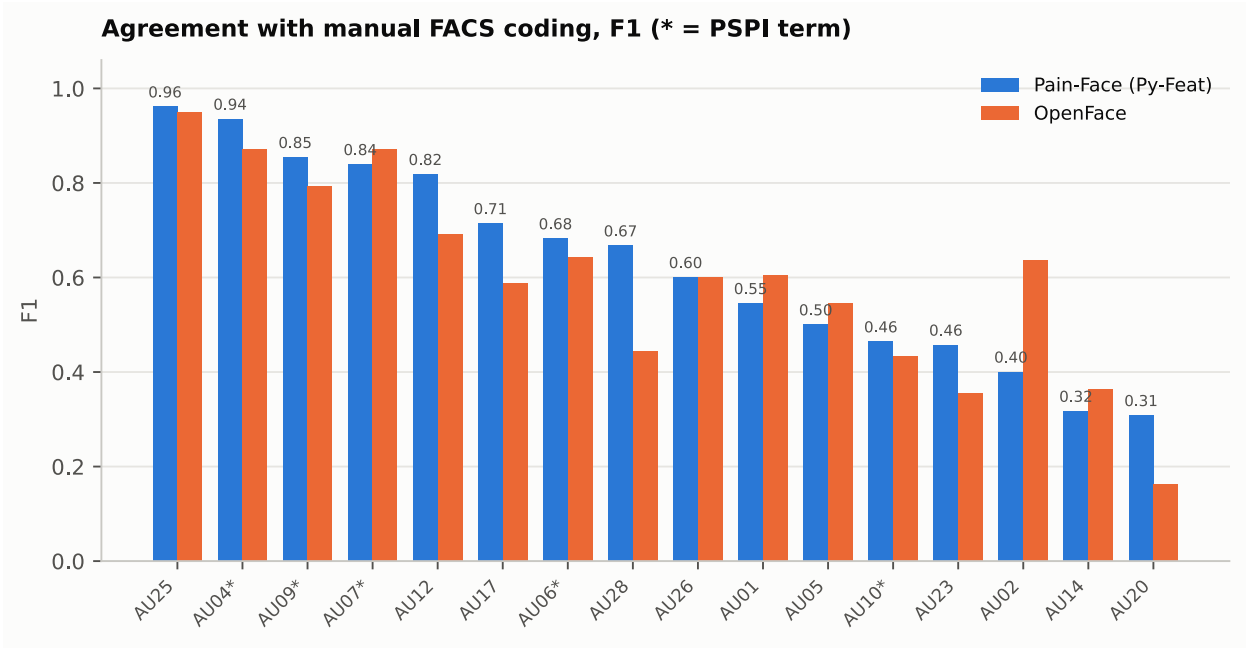


Figure 2: Per-AU agreement with manual FACS coding (F1, threshold 0.5). Pain-Face exceeds OpenFace on 10 of 16 comparable AUs. Asterisks mark the AUs that enter the PSPI formula. AU10 is poor for both tools.

Mean F1 across the 16 AUs with sufficient variance is **0.629** for Pain-Face against **0.597** for OpenFace; mean Cohen's κ is **0.481** against **0.381**. Restricted to the five AUs entering PSPI, Pain-Face mean F1 is **0.755**.

Action unit	Prevalence	F1 (ours)	κ (ours)	F1 (OpenFace)
AU25 lips part	0.547	**0.961**	0.915	0.950
AU04 brow lowerer*	0.853	**0.935**	0.661	0.872
AU09 nose wrinkler*	0.547	**0.854**	0.707	0.793
AU07 lid tightener*	0.768	0.839	0.351	0.870
AU12 lip corner puller	0.368	**0.818**	0.722	0.692
AU17 chin raiser	0.211	**0.714**	0.633	0.588
AU06 cheek raiser*	0.379	**0.682**	0.436	0.643
AU28 lip suck	0.074	**0.667**	0.638	0.444
AU26 jaw drop	0.221	0.600	0.539	0.600
AU01 inner brow raiser	0.221	0.545	0.458	0.604
AU05 upper lid raiser	0.053	0.500	0.479	0.545
AU10 upper lip raiser*	0.274	0.464	0.076	0.433
AU23 lip tightener	0.158	0.457	0.338	0.355
AU02 outer brow raiser	0.074	0.400	0.372	0.636
AU14 dimpler	0.158	0.317	0.101	0.364
AU20 lip stretcher	0.095	0.308	0.265	0.163

* enters the PSPI formula. Bold marks where Pain-Face exceeds OpenFace. Rows in red are below an F1 of 0.5 and should not be relied upon.

AU10 is the significant negative result. At F1 0.464 and κ 0.076 its detection is barely above chance, and it is one half of the PSPI levator term. OpenFace performs no better (F1 0.433, κ 0.054), which suggests an intrinsically difficult AU rather than a defect specific to our pipeline — but it directly caps the reliability of that term, and by extension of PSPI itself. Because the term takes $\max(\mathrm{AU9}, \mathrm{AU10})$ and AU9 is detected well (F1 0.854), the practical damage is limited, but it is real.

9.3 Correlation with observer pain ratings

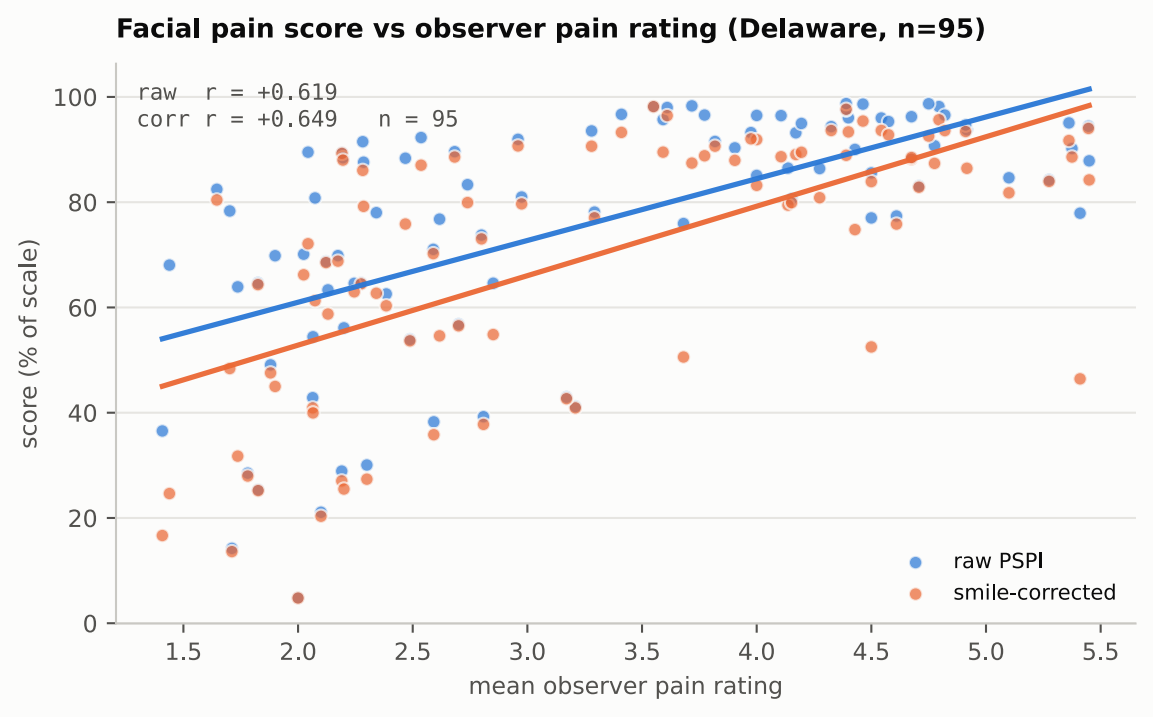


Figure 3: Facial pain score against mean observer pain rating. Both raw and corrected scores correlate significantly; the correction improves the relationship modestly and consistently.

Score	Pearson r	p	Spearman ρ	n
Raw PSPI	+0.619	2.3×10^{-11}	+0.650	95
Smile-corrected	**+0.649**	1.2×10^{-12}	**+0.664**	95

This is the central validation result for the facial channel: the score tracks human judgements of pain intensity with a moderate-to-strong, highly significant association, and the specificity correction improves rather than degrades it.

9.4 Does the specificity correction work?

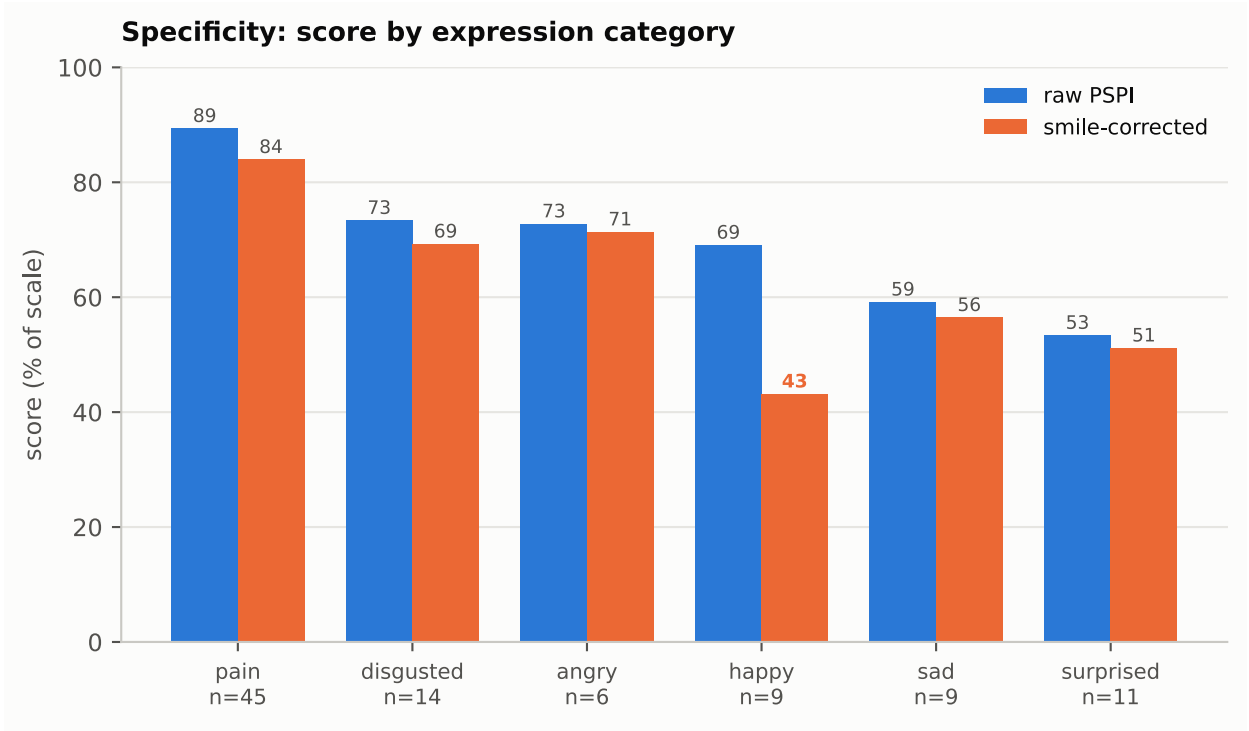


Figure 4: Mean score by dominant expression category, before and after the specificity correction. The correction acts almost entirely on **happy**, leaving pain nearly untouched — and leaving disgust and anger untouched also, which is the system's principal remaining confound.

On the 9 images whose dominant expression is **happy**, raw PSPI reads 11.04 / 16 (69.0% of scale). The corrected score reads **43.1%** — a 25.9-point reduction. On the 45 **pain** images the same correction moves 89.3% to 83.9%, a reduction of only 5.4 points. Mean smile evidence is 0.640 on happy faces against 0.123 on pain faces.

The correction therefore discriminates rather than uniformly attenuating, which is the claim it was designed to support, and it does so on real human faces rather than on the single bundled sample that motivated it.

9.4.1 Where it does not help

Disgust (73% → 69%) and anger (73% → 71%) are barely affected, because the correction keys on AU12 and happiness, neither of which those expressions exhibit. They share AU4, AU9 and AU10 with pain through a different route. Against a mean of 84% for pain, disgust at 69% is not comfortably separated.

Consequently, categorical separation of pain from all other expressions **slightly decreases** under correction: AUC **0.846** → **0.828**. The correction improves agreement with rated intensity while marginally reducing the pain-versus-rest decision boundary. We report both directions rather than selecting the favourable one. For an application whose purpose is graded intensity estimation the trade is defensible; for a binary alarm it may not be.

9.5 Demographic breakdown

Target identifiers in the database encode self-reported race and sex, permitting a stratified analysis — a dimension absent from our prior study.

Group	n	Pain-AU F1	r (score vs rating)	Face quality
Asian	24	0.686	0.577	1.00
Black	23	0.780	0.550	1.00
Latinx	23	0.754	0.616	1.00
White	23	**0.781**	**0.736**	1.00
Female	46	0.783	0.661	1.00
Male	47	0.733	0.598	1.00

Face detection succeeded uniformly. AU detection quality varies by a spread of **0.095** in F1 across race groups, lowest for Asian targets; correlation with observer ratings varies more widely, from 0.550 (Black) to 0.736 (White). Group sizes are small (n ≈ 23–24) and these differences are not tested for significance, so they should be read as an indication that stratified evaluation is necessary, not as an established effect size. We report them because omitting them would misrepresent the system's readiness.

10 Characterisation of the implementation

These experiments establish the operating envelope. They are measurements of system behaviour, not evidence about pain.

10.1 Pulse estimation against signal quality

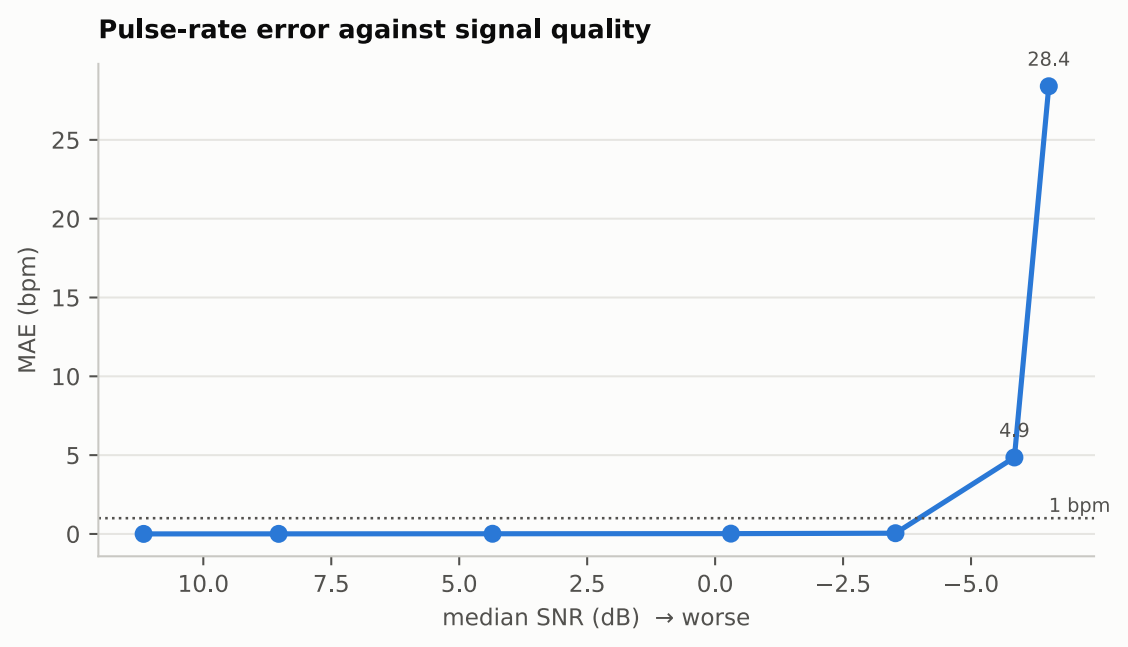


Figure 5: Fused heart-rate error against signal quality, averaged over 54/72/96/120 bpm synthetic traces. Accurate to below 0.1 bpm while SNR exceeds roughly –3.5 dB; collapse is abrupt below that.

Error remains under 0.05 bpm down to a median SNR of –3.5 dB, then degrades sharply: 4.85 bpm at –5.8 dB and 28.4 bpm at –6.5 dB. The transition is a cliff rather than a slope, which is why a reported SNR accompanies every estimate in the interface.

10.2 The abstention boundary

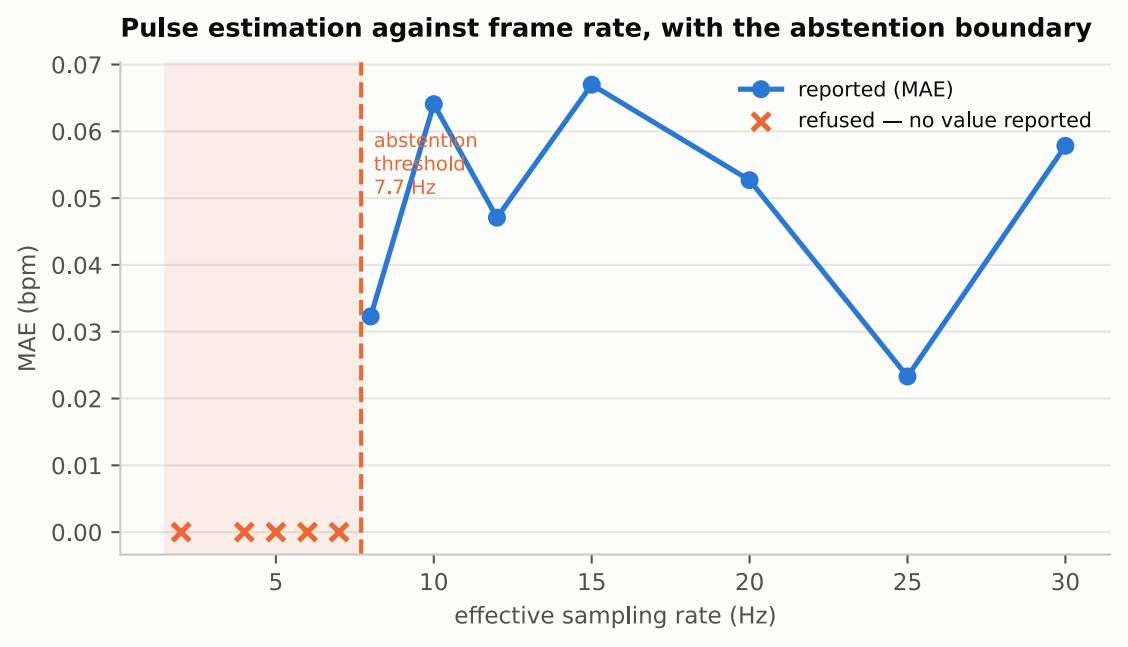


Figure 6: Pulse estimation against effective frame rate. Above the 7.7 Hz threshold, error stays below 0.1 bpm; below it the system reports nothing rather than reporting an aliased value.

At 8 Hz and above, all four test rates are accepted with MAE ≤ 0.07 bpm. At 7 Hz and below, every estimate is refused. The threshold is therefore neither over-conservative (8 Hz remains accurate) nor permissive.

10.3 Method bank

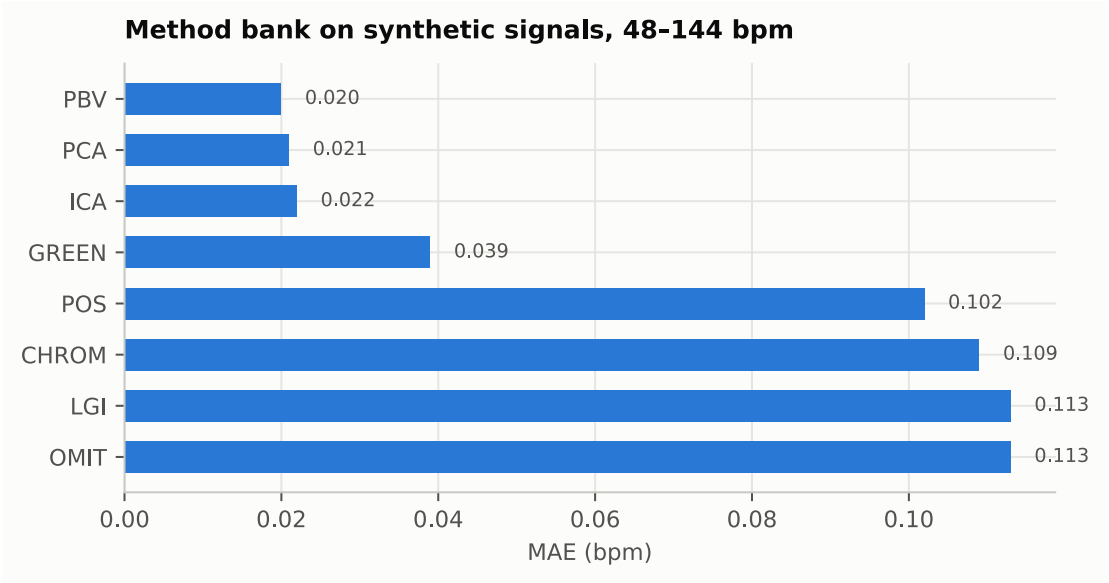


Figure 7: Per-algorithm error on clean synthetic signals across 48–144 bpm.

All eight algorithms resolve clean synthetic signals to better than 0.12 bpm MAE, ranging from PBV (0.020) to LGI and OMIT (0.113). **This ordering should not be read as a general ranking**: the synthetic generator imposes a fixed blood-volume colour direction, which structurally favours methods that estimate that direction (PBV, PCA, ICA) over those using fixed projections tuned for real skin under real illumination (CHROM, POS). Ranking these algorithms requires recorded video with reference PPG.

10.4 Fusion response

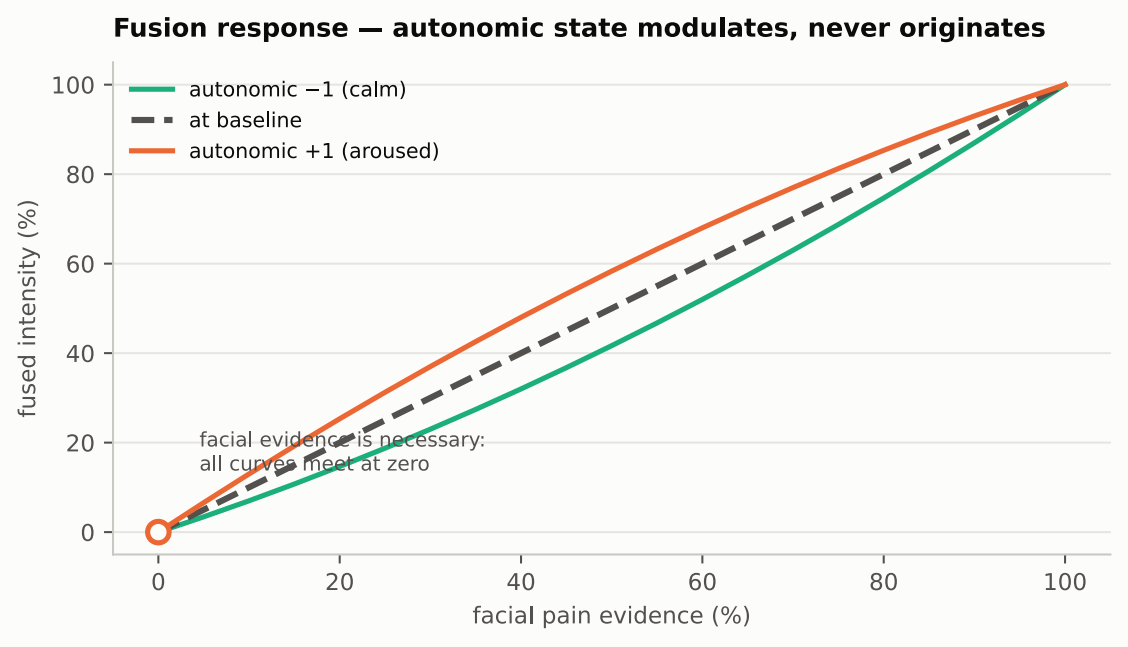


Figure 8: Fused intensity as facial evidence varies, at three autonomic states. All three curves meet at the origin: facial evidence is necessary.

Across the full input grid the maximum attainable intensity is 1.0 and the value at zero facial evidence is exactly 0.0 for every autonomic state, confirming the two properties claimed in §7.2.

10.5 Detector throughput

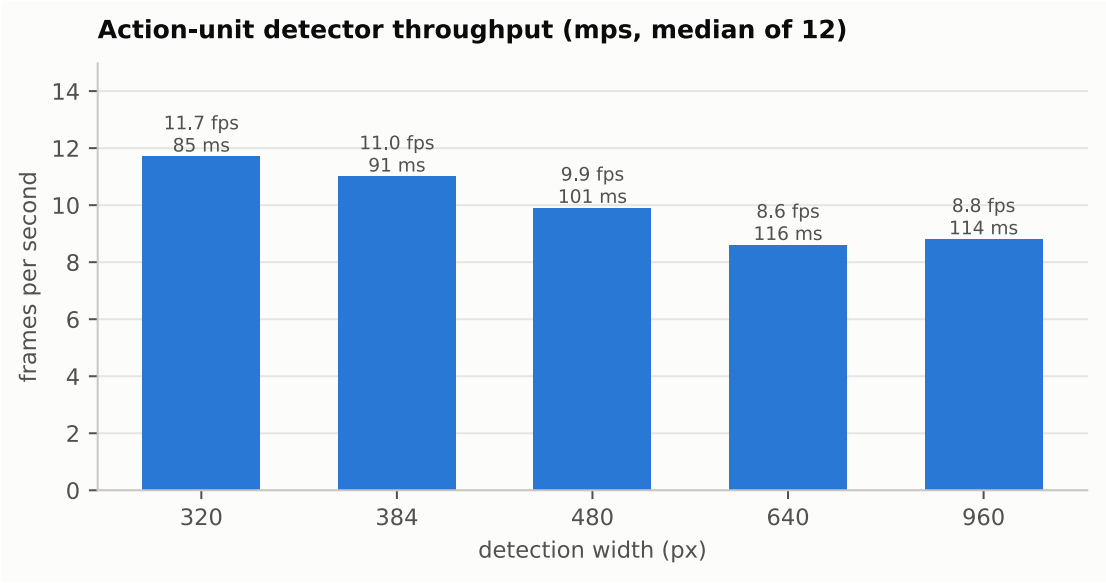


Figure 9: Action-unit inference throughput by detection width on Apple MPS, median of 12 runs after warm-up.

Median latency at 480 px is 100.7 ms (9.9 fps), rising to 11.7 fps at 320 px. **These are medians, not best-case figures** a best-of-N measurement on the same hardware reports values near 17 fps, which we consider unrepresentative of sustained operation. Throughput is the constraint that motivates the dual-rate architecture of §4.1.

10.6 Trigger timing

Onset and release latency equal the configured dwell and release periods exactly across all four tested configurations (0.3/0.8, 0.6/1.5, 1.0/2.5, 2.0/4.0 s), confirming deterministic behaviour with no additional lag.

11 Verified versus validated

We separate these deliberately.

Component	Verified	Validated
PSPI arithmetic	Yes — hand-worked cases	n/a (definition)
AU detection	Yes	**Yes** — vs human FACS, n=95
Specificity correction	Yes	**Yes** — vs expression category
Facial score vs pain	Yes	**Yes** — r = 0.649 vs observer ratings
rPPG algorithms	Yes — synthetic	No
HRV / autonomic features	Yes — synthetic	No
Baseline normalisation	Yes — unit tests	No
Fusion model	Yes — properties proven	No
Episode trigger	Yes — timing exact	No
Abstention rules	Yes — boundary measured	n/a

No claim is made that the fused multimodal index measures pain. The facial component correlates with human pain judgements on still images. The autonomic component and the fusion that combines them have never been tested against any ground truth, because doing so requires synchronised video, reference physiology, and pain labels — data we do not yet have.

The implementation is covered by **91 automated tests** across 4,749 lines of code, pinning the PSPI formula term by term, the fusion invariants, trigger hysteresis under a non-monotonic clock, heart-rate recovery at four rates, the abstention boundary, and the full WebSocket protocol.

12 Limitations

1. **Observable behaviour, not experience.** People mask pain; stoicism reads as absence. A low score is not evidence of comfort.
2. **Probabilities are not intensities.** Py-Feat reports occurrence probability; PSPI is defined on A–E intensities. Our mapping preserves the scale's range and ordering but not its coder-equivalent semantics.
3. **AU10 is unreliable** (F1 0.464), weakening one half of the levator term.
4. **Disgust and anger are confounded with pain** and the present correction does not address them. This is the most important open problem.
5. **The specificity correction trades categorical separation for intensity correlation** (AUC 0.846 → 0.828).
6. **Still images only.** All validation here is on posed and evoked stills from one database. Nothing establishes behaviour on continuous video, spontaneous clinical pain, or patients.
7. **Demographic performance varies** across the groups measured, with small n.
8. **The autonomic channel is unvalidated**, and its features are not pain-specific: exercise, anxiety, caffeine and ambient temperature all move them. This is why the fusion lets them modulate but never originate a reading.
9. **rPPG is fragile**: it requires diffuse flicker-free illumination, a reasonably still subject, and locked exposure. Auto-exposure injects noise directly into the cardiac band.
10. **Fusion weights are reasoned, not fitted.** $W_f = 0.75$, $W_a = 0.25$, the discount factors and the trigger thresholds derive from the structure of the literature, not from optimisation against labelled data. Their **form** is defended in §7.2 and verified; their **values** are a starting point.

13 Proposed validation protocol

The following would convert the unvalidated rows of §11 into measured ones.

Stage 1 — Video facial validation (UNBC-McMaster). The UNBC-McMaster Shoulder Pain Expression Archive [16] provides 200 sequences from 25 patients with frame-level PSPI coded by certified FACS coders and self-report. Required outputs: per-frame MAE and RMSE against coder PSPI; Pearson and Spearman

correlation; trigger ROC against sequence-level pain; per-patient breakdown. Requires a signed academic data agreement.

Stage 2 — Autonomic validation. Synchronised recording of face video with a reference contact PPG or ECG, under rest and a controlled nociceptive stimulus (for example cold-pressor or pressure algometry, with ethical approval). Required outputs: Bland–Altman agreement of rPPG heart rate against reference; RMSSD and LF/HF agreement; sensitivity of each autonomic feature to the stimulus.

Stage 3 — Fusion validation. With Stages 1 and 2 in place, test whether the fused index outperforms the facial channel alone against self-reported pain — the central unproven claim of this work. An ablation (facial only, autonomic only, fused) is the minimum required analysis.

Stage 4 — Fitting and fairness. Replace the reasoned constants with values fitted on held-out data, and report performance stratified by skin tone under the Monk or Fitzpatrick scale, which the present data does not support.

14 Reproducibility

Source: `https://github.com/NaveedShariff/Pain-Face`.

Component	Version
Python	3.11.16
py-feat	2.1.3 (`Detectorv2`, device `mps`)
PyTorch	2.14.1
NumPy / SciPy	2.4.6 / 1.17.1
OpenCV	5.0.0

Every number in this report regenerates from the repository:

```
python -m pytest                      # 91 tests
python validation/run_delaware.py     # score the database
python validation/analyse.py          # agreement, correlation, specificity
python validation/characterise.py     # the six characterisation experiments
python validation/figures.py          # all figures
```

Ground truth is read directly from the Delaware distribution's `DelawarePainDatabase_StimulusCharacterization_forOSF.xlsx`; no labels were re-derived or hand-adjusted.

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